

1. Use the quadratic formula to solve the equation.

$$x^2 + 3x - 8 = 0$$

x =

(Simplify your answer. Type an exact answer, using radicals as needed. Use a comma to separate answers as needed.)

2. Give exact and approximate solutions to three decimal places.

$$5x^2 = 3 + 3x$$

The approximate solution(s) is/are $x \approx$.

(Use a comma to separate answers as needed. Round to the nearest thousandth as needed.)

3. The U.S. Crude Oil production, in billions of barrels, for the years from 2015 projected to 2035, can be modeled $y = -0.002x^2 + 0.052x + 1.954$, with x equal to the years after 2015 and y equal to the number of billions of barrels of crude oil.

- Find and interpret the vertex of the graph of this model.
- What does the model predict the crude oil production will be in 2038?
- Graph the function for the years 2015 to 2035.

- a. The vertex of the graph of this model is $v = (\text{ }, \text{ })$. (Round to three decimal places as needed.)

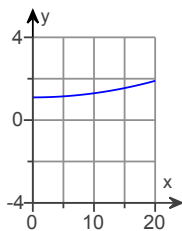
Choose the correct interpretation of the vertex below.

- ☐ The maximum number of barrels of crude oil projected to be produced during this period is 13.000 billion barrels.
- ☐ The minimum number of barrels of crude oil projected to be produced during this period is 2.292 billion barrels.
- ☐ The maximum number of barrels of crude oil projected to be produced during this period is 2.292 billion barrels.
- ☐ The minimum number of barrels of crude oil projected to be produced during this period is 13.000 billion barrels.

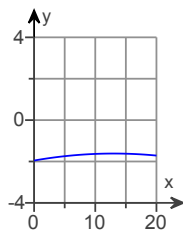
- b. The model predicts the crude oil production to be billion barrels of oil in 2038. (Round to three decimal places as needed.)

- c. Choose the correct graph below.

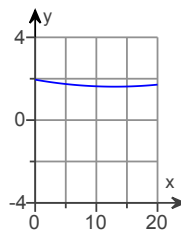
☐ A.



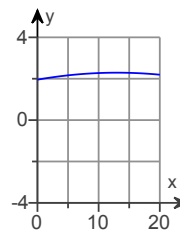
☐ B.



☐ C.



☐ D.



*4.

Suppose that a person's heart rate x minutes after vigorous exercise has stopped can be modeled by $f(x) = \frac{5}{6}(x - 12)^2 + 70$. The output is in beats per minute, where the domain of f is $0 \leq x \leq 12$. **(a)** Evaluate $f(0)$ and $f(3)$. Interpret the result. **(b)** Estimate the times when the person's heart rate was between 120 and 150 beats per minute, inclusive.

(a) $f(0) =$ and $f(3) =$
(Simplify your answer. Type an integer or decimal.)

Choose the correct interpretation.

- ☐ **A.** Rate is 190 bpm initially and about 138 bpm after 3 min.
☐ **B.** Rate is 138 bpm initially and about 190 bpm after 3 min.

(b) The times when the person's heart rate was between 120 and 150 beats per minute is about $\leq x \leq$ (min).
(Simplify your answer. Type an integer or decimal rounded to one decimal place as needed.)

5. How is the graph of $y = (x + 1)^2 + 2$ transformed from the graph of $y = x^2$?

Choose the correct answer below.

- ☐ **A.** horizontal shift 1 unit to the left, vertical shift 2 units up
☐ **B.** horizontal shift 1 unit to the left, vertical shift 2 units down
☐ **C.** horizontal shift 1 unit to the right, vertical shift 2 units down
☐ **D.** horizontal shift 1 unit to the right, vertical shift 2 units up

6. The number of millions of people age 12 and older who used marijuana during the years 2003 to 2008 is described by the function $M(x) = -0.066(x - 4.9)^2 + 25.8$ for $3 \leq x \leq 8$, where x is the number of years after 2000.

- a.** The graph of this function is a shifted graph of which basic function?
b. Find and interpret $M(5)$.
c. Sketch a graph of $y = M(x)$ for $3 \leq x \leq 8$.

a. Choose the correct answer below.

- ☐ $f(x) = x^2$
☐ $f(x) = x^3$
☐ $f(x) = \sqrt{x}$
☐ $f(x) = |x|$

b. Find $M(5)$.

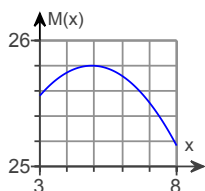
$M(5) =$

(Type an integer or a decimal rounded to three decimal places as needed.)

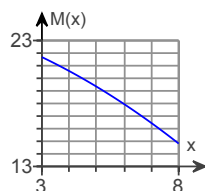
This means that in , the number of people 12 and older who used marijuana was .

c. Choose the correct graph of the function for $3 \leq x \leq 8$ below

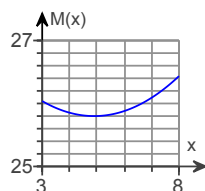
☐ **A.**



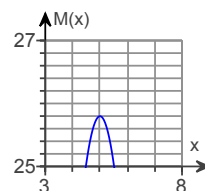
☐ **B.**



☐ **C.**



☐ **D.**



7. The total revenue function for a product is given by $R = 745x$ dollars, and the total cost function for this same product is given by $C = 10,500 + 30x + x^2$, where C is measured in dollars. For both functions, the input x is the number of units produced and sold.
- Form the profit function for this product from the two given functions.
 - What is the profit when 13 units are produced and sold?
 - What is the profit when 20 units are produced and sold?
 - How many units must be sold to break even on this product?

- a. Write the profit function.

$P(x) =$

(Simplify your answer.)

- b. When 13 units are produced and sold, there is a

(1) _____ of \$.

(Simplify your answer.)

- c. When 20 units are produced and sold, there is a

(2) _____ of \$.

(Simplify your answer.)

- d. The number of units that must be sold to break even on this product is .

(Simplify your answer. Use a comma to separate answers as needed.)

- (1) ☐ loss (2) ☐ profit
☐ profit ☐ loss